

KEY2USB KIT

Use your Commodore 64's own keyboard as a USB keyboard for VICE — no soldering into the machine, no case modification.

Thank you for your order! Everything you need is in this box: the PCB, all components, and a 3D-printed case. **The two programmed chips — the 28C64 EEPROM (U1) and the ATmega328 (U5) — arrive pre-programmed and ready to run**, so there is nothing to flash. All you need to supply is a soldering iron, solder, and side cutters. If anything is missing or damaged, email me and I'll make it right.

 **SET VICE TO POSITIONAL**

Settings → Keyboard → Keyboard Mapping → Positional — not Symbolic. This is the single most common first-time mistake; if your keys come out wrong, check it before anything else. KEY2USB reports keys by *physical position*, never as characters. Symbolic layers an unwanted PC-layout translation on top of that and garbles shifted keys — SHIFT + 2 arrives as @ instead of ".

WHAT'S IN THE KIT

QTY	REF	PART
1	—	KEY2USB PCB + 3D-printed case
1	U1	28C64 EEPROM + socket — pre-programmed
1	U5	ATmega328 MCU + socket — pre-programmed
3	U2–U4	74LS32 (DIP-14), 74LS273 (DIP-20), 74LS74 (DIP-14) glue logic
3	Y1, C5, C6	16 MHz crystal + 2× 20pF load caps
4	C1–C4	100nF ceramic — IC bypass
2	D1, D2	1N5227B 3.6V Zener — <i>polarized</i>
3	R1–R3	2× 68Ω USB series, 1× 1.5kΩ D– pull-up
2	J1, J3	3-pin MODE header + jumper cap, USB Type B connector

ASSEMBLY

All through-hole. Build shortest-to-tallest so parts sit flat. The two programmed chips get **sockets**; the three 74LS logic ICs solder **directly** down.

- Resistors **R1–R3**, then ceramic capacitors **C1–C6** — neither is polarized.
- Zener diodes **D1 D2** — **polarized**. The band on the body must match the banded end of the silkscreen outline.
- The two DIP-28 sockets at **U1** and **U5**, notch matching the silkscreen. Leave them empty for now.
- U2 U3 U4** soldered straight down — **check each notch against the silkscreen first**; these are not socketed and are unpleasant to remove.
- Crystal **Y1**, MODE header **J1**, then the USB Type B connector **J3** — seat J3 fully flat; it takes the mechanical stress of the cable.
- Trim all leads flush, then inspect for solder bridges — especially along the edge fingers and between adjacent IC pins.
- Press the labeled **U1** and **U5** chips into their sockets last, notch matching, with no bent-under pins.

Before closing the case: fit the jumper cap on **J1** in the C64 position (the GND side, marked on the silkscreen) — the supported setting, and covers every machine listed below. (The board's other firmware bank holds an unsupported C128 native-mode build, kept for the curious.)

COMPATIBILITY

Commodore 64 / 64C	Yes — primary target
Commodore 128 / 128D	Yes — boots straight into C64 mode
Ultimate 64	Yes — verified on hardware

On a C128 the cartridge asserts **/EXROM**, so it comes up in C64 mode on its own; if one does not autostart, power on holding **C=**. Its extra keys (keypad, ESC, TAB, ALT, HELP, cursor pad) sit outside the standard 8×8 matrix and are not reported; every key it shares with the C64 works.

FIRST POWER-ON

- With the C64 **powered off**, insert KEY2USB into the expansion port.
- Connect the USB cable to your PC. It enumerates as a standard HID keyboard — no drivers needed.
- Power on the C64. The **KEY2USB** splash screen appears; the cartridge has taken over the machine and is scanning the keyboard.
- In VICE, set **Settings → Keyboard → Keyboard Mapping → Positional**. This step is mandatory.
- Type on the C64. Keystrokes arrive at the PC.

While KEY2USB is running the C64 is dedicated to being a keyboard and does not return to BASIC. Power off and remove the cartridge to use the machine normally.

JOYSTICKS IN VICE

Both joystick ports are scanned and arrive at the host as ordinary keypad keystrokes, so VICE must be told which keys mean "joystick."

Port 2 — no configuration needed beyond **Settings → Joystick → Joystick Port #2 → Numpad**. VICE's built-in Numpad device already matches the cartridge's port-2 mapping exactly.

Port 1 — needs a keyset, since Numpad can only serve one port. Set **Joystick Port #1 → Keyset A**, then **Configure Keyset A**:

DIRECTION	KEY
North	Keypad –
South	Keypad +
West	Keypad /
East	Keypad *
Fire	Keypad enter

Worth knowing: a held joystick direction grounds a line in the keyboard matrix, so it phantom-presses real keys — exactly as on a bare C64. Port 1 is hit hardest, grounding a whole matrix row (RETURN, INST/DEL, CLR/HOME), so expect stray keystrokes while a port-1 direction is held. **Port 2 is the one to use.**

TROUBLESHOOTING

SYMPTOM	CHECK
Wrong characters	VICE keyboard mapping is set to Positional , not Symbolic.
No splash screen	Cartridge fully seated, C64 powered off when inserted. Then re-check IC orientation and bridged joints near U1 and the edge fingers.
No USB device	The C64 must be powered on — the cartridge is bus-powered. Then check Y1 and its 20pF caps, R1–R3, and the D1/D2 band orientation.

Joystick does nothing Port 2 set to **Numpad**; port 1 set to **Keyset A** and configured as above.

`Software/key2usbttest.prg` in the repository shows live joystick state and the last keys received — the quickest way to check a finished build.